

Kyubin James Kwon

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EDUCATION	University of California, Santa Barbara	2021–2027 (expected)
	Ph.D. Physics (expected)	
	M.A. Physics	2024
	Advisor: Prof. Tejaswi Venumadhav	
	University of California, Berkeley	2021
RESEARCH INTERESTS	High Distinction, B.A. Honors Astrophysics; B.A. Physics	
	Thesis: <i>deepCR on ACS/WFC: Cosmic-Ray Rejection for HST ACS/WFC Photometry</i>	
	Advisor: Prof. Joshua Bloom	
	Tidal interactions, neutron stars, gravitational waves, general relativity, asteroseismology, statistical methods	
INVITED TALKS & SEMINARS	University of Birmingham Astrophysics Seminar, <i>Nonlinear Tides in Coalescing BNS</i>	2025
	ICTS Astrophysics & Relativity Seminar, <i>Nonlinear Tides in Coalescing Neutron Star Binaries</i>	2024
	Montana State University Gravity Group Seminar, <i>Nonlinear Tidal Effects on BNS Inspiral</i>	2024
CONTRIBUTED TALKS & POSTERS	APS Global Physics Summit, <i>Resonance Locking of g-Modes in Coalescing Neutron Star Binaries</i>	2026
	APS Global Physics Summit, <i>General Relativistic Triplet Interaction of Oscillation Eigenmodes in Deformed Neutron Stars</i>	
	GWPAW25, <i>Resonance Locking of g-Modes in Coalescing Neutron Star Binaries</i>	2025
	The 41 st Pacific Coast Gravity Meeting, <i>Resonance Locking of g-Modes in Neutron Star Binaries</i>	
	APS Global Physics Summit, <i>Resonance Locking of g-Modes in Coalescing Neutron Star Binaries</i>	
	UCSB AstroLunch, <i>Resonance Locking of g-Modes in Coalescing Neutron Star Binaries</i>	2024
	GWPAW24, <i>Nonlinear Resonance Locking in Coalescing Binary Neutron Stars</i>	
	SBI for Galaxy Evolution, <i>Normalizing Flow to Model the Kinematics of Galaxies in Halos</i>	
	The 40 th Pacific Coast Gravity Meeting, <i>Tidal pg-instability in Coalescing Binary Neutron Stars</i>	
	UCSB AstroTea, <i>Neural Stellar Population Synthesis Emulator for Probabilistic VAC</i>	2023
	UCSB Astro Lunch, <i>The Probabilistic Value-Added Catalog for Bright Galaxy Survey</i>	2022
	The 54th Sierra Conference , <i>The Probabilistic Value-Added BGS Catalog</i>	
	AAS237 , <i>deepCR on ACS/WFC: Cosmic-Ray Rejection Model for HST ACS/WFC Photometry</i>	2021
	DESI Research Forum, <i>GQP Mock Challenge for Probabilistic Value-Added BGS Catalog</i>	2020

AWARDS, HONORS & GRANTS	Individualized Professional Skills Grant, UCSB	2026
	Academic Senate Doctoral Student Travel Grant, UCSB	
	Academic Senate Doctoral Student Travel Grant, UCSB	2025
	Graduate Student Association Travel Grant, UCSB	
	Phi Beta Kappa, UC Berkeley	2021
	Chambliss Astronomy Award Honorable Mention, AAS 237	
	Dean’s List, College of Letters & Science, UC Berkeley	
RESEARCH EXPERIENCE	Academic Visitor, <i>University of Birmingham, United Kingdom</i>	2025
	Academic Visitor, <i>International Centre for Theoretical Sciences, India</i>	2023, 2024
	Graduate Student Researcher, <i>UC Santa Barbara, USA</i>	2022–
	Research Intern, <i>NASA Ames & KARI¹, USA</i>	Postponed
	Undergraduate Research Apprentice Program, <i>UC Berkeley, USA</i>	2019–2021
TEACHING & MENTORING	Course Reader (Statistical Mechanics), UCSB	2026
	Instructional Laboratory Group, UCSB	2025
	Lead Teaching Assistant (Physics Laboratory), UCSB	2022
	Teaching Assistant (Physics Laboratory), UCSB	2021
	Course Reader (Relativistic Astrophysics and Cosmology), UC Berkeley	2021
	Course Reader (Stellar Physics), UC Berkeley	2020
	Math Tutor, Diablo Valley College	2018–2019
TECHNICAL SKILLS	<i>Languages:</i> Python, C++, Shell scripting <i>Methods:</i> High-Performance Computing (HPC); Parallel Computing (MPI, OpenMP); Machine Learning (TensorFlow, PyTorch); Agentic Coding (Claude) <i>HPC Machines:</i> UCSB HPC Cluster; National Energy Research Scientific Computing Center–Cori (2M+ CPU hours); Exalearn (Lawrence Berkeley National Laboratory)	
SERVICE & OUTREACH	Referee, ApJS	2026
	UCSB AstroTea Organizer	2023–
	Invited Writer, Astrobites	2021
	Language Mentor for War-Impacted Ukrainian Students	2023–2026
	Co-chair of Astrophysics Departmental Undergraduate Task Force, UC Berkeley	2020–2021
PUBLICATIONS & PREPRINTS	[†] : first author; bold <i>et al.</i> : name among the omitted co-authors. 215 citations, <i>h</i> -index 9 (ADS, September 2026). Neutron-Star Tides & Gravitational Waves Kwon, K.J.[†] , <i>et al.</i> (in prep), <i>Nonlinear Perturbations of Relativistic Stars</i> Kwon, K.J.[†] , <i>et al.</i> (in prep), <i>Adiabatic Nonlinear Oscillations of Non-Relativistic Superfluid Stars</i> Yu, H., <i>et al.</i> 2026, arXiv e-prints, doi: 10.48550/arXiv.2607.07943 , <i>Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions</i>	

¹Korea Aerospace Research Institute; the internship was postponed after the NASA Ames Center closed due to COVID-19.

- Hegade K.R., A., **Kwon, K.J.**, Venumadhav, T., Yu, H., & Yunes, N. 2026, arXiv e-prints, doi: [10.48550/arXiv.2605.08569](https://doi.org/10.48550/arXiv.2605.08569), *The Good, the Bad, and the Subtle: Relativistic mode sums for neutron-star tidal response*
- Hegade K.R., A., **Kwon, K.J.**, Venumadhav, T., Yu, H., & Yunes, N. 2026, PRL, 136, 071401, doi: [10.1103/1wdp-6x27](https://doi.org/10.1103/1wdp-6x27), *Relativistic and Dynamical Love Numbers*
- Kwon, K.J.**[†], Yu, H., & Venumadhav, T. 2025, arXiv e-prints, doi: [10.48550/arXiv.2503.11837](https://doi.org/10.48550/arXiv.2503.11837), *Resonance locking: radian-level phase shifts due to nonlinear hydrodynamics of g -modes in merging neutron star binaries*
- Kwon, K.J.**[†], Yu, H., & Venumadhav, T. 2024, arXiv e-prints, doi: [10.48550/arXiv.2410.03831](https://doi.org/10.48550/arXiv.2410.03831), *Resonance Locking of Anharmonic g -Modes in Coalescing Neutron Star Binaries*
- Yu, H., Weinberg, N.N., Arras, P., **Kwon, J.**, & Venumadhav, T. 2023, Monthly Notices of the Royal Astronomical Society, 519, 4325, doi: [10.1093/mnras/stac3614](https://doi.org/10.1093/mnras/stac3614), *Beyond the linear tide: impact of the non-linear tidal response of neutron stars on gravitational waveforms from binary inspirals*

Statistical Methods in Astrophysics & Cosmology

- Agarwal, S., *et al.* 2026, ApJ, 1001, 162, doi: [10.3847/1538-4357/ae4a9a](https://doi.org/10.3847/1538-4357/ae4a9a), *DESI Strong Lens Foundry. III. Keck Spectroscopy for Strong Lenses Discovered Using Residual Neural Networks*
- Huang, X., *et al.* 2026, ApJ, 998, 69, doi: [10.3847/1538-4357/ae22d2](https://doi.org/10.3847/1538-4357/ae22d2), *DESI Strong Lens Foundry. I. HST Observations and Modeling with GIGA-Lens*
- Huang, X., *et al.* 2025, arXiv e-prints, doi: [10.48550/arXiv.2509.18089](https://doi.org/10.48550/arXiv.2509.18089), *DESI Strong Lens Foundry II: DESI Spectroscopy for Strong Lens Candidates*
- Kwon, K.J.**[†], & Hahn, C. 2024, ApJ, 976, 76, doi: [10.3847/1538-4357/ad8442](https://doi.org/10.3847/1538-4357/ad8442), *Investigating the Kinematics of Central and Satellite Galaxies Using Normalizing Flows*
- Storfer, C., *et al.* 2024, ApJS, 274, 16, doi: [10.3847/1538-4365/ad527e](https://doi.org/10.3847/1538-4365/ad527e), *New Strong Gravitational Lenses from the DESI Legacy Imaging Surveys Data Release 9*
- Kwon, K.J.**[†], Hahn, C., & Alsing, J. 2023, ApJS, 265, 23, doi: [10.3847/1538-4365/acba14](https://doi.org/10.3847/1538-4365/acba14), *Neural Stellar Population Synthesis Emulator for the DESI PROVABGS*
- Hahn, C., **Kwon, K.J.**, *et al.* 2023, ApJ, 945, 16, doi: [10.3847/1538-4357/ac8983](https://doi.org/10.3847/1538-4357/ac8983), *The DESI PRObabilistic Value-added Bright Galaxy Survey (PROVABGS) Mock Challenge*

NON-REFEREED PUBLICATIONS

- Lovell, C.C., *et al.* 2023, in Machine Learning for Astrophysics, 21, doi: [10.48550/arXiv.2307.06967](https://doi.org/10.48550/arXiv.2307.06967), *A Hierarchy of Normalizing Flows for Modelling the Galaxy-Halo Relationship* (workshop proceedings)
- Kwon, K.J.**[†], Zhang, K., & Bloom, J.S. 2021b, RNAAS, 5, 98, doi: [10.3847/2515-5172/abf6c8](https://doi.org/10.3847/2515-5172/abf6c8), *deepCR on ACS/WFC: Cosmic-Ray Rejection for HST ACS/WFC Photometry* (research note)
- Kwon, K.J.**[†], Zhang, K., & Bloom, J.S. 2021a, in AAS Meeting Abstracts, Vol. 237, AAS Meeting Abstracts, 350.02, *deepCR-ACS/WFC: Cosmic-Ray Rejection Model for HST ACS/WFC Photometry* (meeting abstract)

REFERENCES

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